BUSINESS CASE / NOVEMBER 2026

Corporate Financial Forecasting & Performance Analysis

An end-to-end, evidence-driven financial modeling and decision-support portfolio project

PROPOSED PROJECT

Build one reusable financial analysis case study from public-company filings: historical financial analysis, an integrated forecast, scenario planning, validated assumptions, an executive dashboard, and a recommendation memo.

FINANCE FIRST

Three financial statements, operating drivers, working capital, and free cash flow.

AUDITABLE MODEL

Source-linked SEC data, explicit assumptions, reconciliation, and backtesting.

ACTIONABLE STORY

Scenario sensitivities, KPI dashboard, and concise management recommendations.

Status: Proposed / not built. This document defines required work, quality gates and deliverables. It does not claim that historical data have been collected, models have been built, or forecast performance has been achieved.

Prepared for a broadly transferable finance-analyst project portfolio; not tailored to one employer, internship, industry, or investment recommendation.

01 / THE INVESTMENT THESIS

Executive business case

Why this project is worth building, what decision it supports, and how success will be judged.

Problem / opportunity

Entry-level finance and strategy roles increasingly require more than descriptive dashboards. They value analysts who can interpret historical performance, explain the business drivers behind a forecast, assess risks, and communicate what decisions follow. A portfolio can demonstrate these capabilities through a reproducible case study even without access to proprietary corporate planning data.

Proposed solution

Create a driver-based, three-statement financial model for **one publicly traded, non-financial operating company**, with 5 fiscal years of historical data where available, a 3-year projection, three operating scenarios, retrospective forecast validation, Power BI reporting, and an executive recommendation memo.

Business-case element	Proposal
Primary user	Hiring managers and interviewers for FP&A, corporate finance, finance/data analyst, banking-adjacent and strategy roles; also the project author as an analyst.
Decision enabled	Which operational or financial drivers should management prioritize under base, upside, and downside conditions?
Evidence created	A sourced, balancing financial model, historical backtests, sensitivity analysis, dashboard, and reproducible files.
Cost / effort	Indicative 45–65 focused hours; public filing data and open-source processing tools can keep direct data costs low. Software licensing may vary.
Desired outcome	A defensible, explainable work sample—not a promise of investment performance or a guaranteed hiring result.

Recommendation & approval request

> **Proceed with an MVP.** Start with annual statements for one suitable company, an integrated model, and one management question. Expand to quarterly forecasting, a second company, or advanced ML only after the core model passes validation.

02 / BUSINESS OBJECTIVES

What the project must prove

Define success as demonstrated analytical competence, not fabricated business impact.

Objective	Observable evidence / acceptance criterion
Financial understanding	Explain trends and drivers for revenue, margins, earnings, capital intensity, working capital and cash generation.
Modeling discipline	Forecast all three statements with linked formulas and a balancing balance sheet across three fiscal years.
Quantitative reasoning	Benchmark retrospective forecasts and document metric definitions, errors, and limitations.
Decision-making	Translate three scenarios and sensitivity results into 2–4 defensible strategic recommendations.
Communication	Provide a 2-page executive memo, a dashboard with defined KPIs, and a 3–5 minute walkthrough.
Reproducibility	Maintain original source links, a data dictionary, scripts/workbook formulas, checks, README and a version history.

Roles and transferable skills

Role family	Evidence that transfers
FP&A / financial analyst	Forecast vs actuals, revenue and expense drivers, cash-flow projection, sensitivity and variance analysis.
Corporate finance	Profitability analysis, working capital, financial statement linkage, capital allocation considerations.
Strategy / consulting	Issue structuring, scenarios, trade-offs, KPI interpretation and management recommendations.
Equity / investment research	Filing research, forecasts and business drivers; optional valuation extension after the core model.
Business / data analyst	ETL, data validation, SQL/Python transformations, visual reporting and documented metrics.

Important: A forecast is an estimate, not proof of financial improvement. Any future resume claim must describe only completed and measured work: data years, number of scenarios, backtest results, and actual analytical outputs.

03 / PROJECT CHARTER

Scope, boundaries & company choice

A deliberately broad finance project with a manageable first implementation.

In scope

- Select one U.S.-listed, non-financial public company with stable disclosure and usable 10-K/10-Q history.
- Document the company and industry context, fiscal-year convention, segments (if disclosed) and key operating drivers.
- Capture and normalize historical annual statements, then model 3 projected fiscal years.
- Build and tie out an income statement, balance sheet, cash-flow statement, driver assumptions and calculated KPIs.
- Run base / upside / downside scenarios, at least two sensitivity tests, and historical backtests.
- Create management-facing dashboards, a written executive memo, and reproducible files.

Out of scope for MVP

- Trading signals, stock-price prediction, algorithmic trading or investment advice.
- A production-grade enterprise planning system or real-time market-data integration.
- Predictive ML that is added only for novelty without a credible baseline.
- Comparing multiple companies before the first model reconciles and validates.

Company selection scorecard

Criterion	Decision rule
Public filings	At least 5 comparable annual reporting periods; audited 10-K statements available.
Business simplicity	Prefer an understandable operating business with visible revenue and cost drivers; avoid banks and insurers for version 1.
Data availability	Core fields available through XBRL or defensibly transcribed from filings, with clear source notes.
Model suitability	Material operating profit and cash-flow items can be projected with reasonable driver logic.
Interview usefulness	Easy to explain business model and key assumptions without company-insider knowledge.

Illustrative candidates: Adobe or Microsoft, subject to a data-completeness check. Choose for clean disclosures and a tractable business model, not because a specific employer was named.

04 / DATA FOUNDATION

Sources, data dictionary & controls

Every meaningful number must have an identifiable source, period, unit, and transformation.

Source	Use and caveat
SEC EDGAR filings (10-K / 10-Q)	Primary audited/periodic statements, MD&A, revenue recognition, segment disclosures, footnotes.
SEC data.sec.gov XBRL APIs	Structured companyfacts and filing histories. Not every company-specific concept appears in standard tags; reconcile with filings.
FRED (optional)	Macroeconomic scenario context, e.g., rates or inflation; disclose series ID, units, date and access method.
Company investor-relations site	Optional presentation or earnings materials used for context; do not substitute unaudited slides for filing reconciliation.

Required minimum data dictionary

Category	Fields / definitions
Provenance	Company, ticker, CIK, form, accession, fiscal period start/end, filing date, source URL, reported unit.
Income	Revenue, COGS, gross profit, operating expenses, EBIT, interest, taxes and net income.
Balance sheet	Cash, receivables, inventory (if relevant), PP&E, other assets, payables, debt, liabilities and equity.
Cash flows	CFO, capex, investing and financing cash flows, depreciation/amortization, net change in cash.
Drivers	Revenue growth, margins, effective tax rate, DSO/DPO/DIO when applicable, capex %, D&A %, financing assumptions.

Data-quality checks

- Preserve raw filing extracts separately from normalized data; never overwrite the original evidence.
- Resolve XBRL concept synonyms and units explicitly; record restatements, amendments and definition changes.
- Do not mix calendar years, fiscal years, year-to-date quarters and stand-alone quarters.
- Verify balance sheet identity, cash-flow reconciliation, sign conventions and major line items against the 10-K.
- Store an as-of date for backtesting so future filings never leak into historical forecast inputs.

Data-source references: SEC EDGAR APIs (sec.gov/search-filings/edgar-application-programming-interfaces); FRED API (fred.stlouisfed.org/docs/api/fred/overview.html). See full links on the final page.

05 / MODEL SPECIFICATION

Integrated three-statement forecast

A rigorous finance model first; use Python and BI tools where they add validation or usability.

Module	Minimum required logic
Operating drivers	Revenue growth or segment drivers; costs/margins; operating expense assumptions; tax rate.
Income statement	Revenue - COGS = gross profit; subtract operating expenses to EBIT; interest and taxes to net income.
Balance sheet	Project operating assets/liabilities, PP&E, debt, cash and equity; link retained earnings to net income and distributions.
Cash-flow statement	Start with net income, adjust non-cash expenses and working capital; subtract capex; reflect debt, dividends and buybacks if material.
Funding / cash loop	Reflect debt and interest assumptions transparently; avoid circular errors or an arbitrary balancing plug.
KPIs	Gross margin, operating margin, net margin, CFO margin, FCF, cash conversion and relevant working-capital ratios.

Linkages to verify

- Net income from the forecast income statement feeds operating cash flow and retained earnings.
- Ending cash equals beginning cash plus net operating, investing and financing flows.
- PP&E; change reflects capex less depreciation and relevant disposals/other adjustments.
- Operating working-capital changes are consistent between projected balance sheet items and CFO.
- Assets = liabilities + equity for every forecast year, within a documented rounding tolerance.

Implementation roles for tools

Tool	Why it is used
Excel	Main transparent financial model: assumptions, statements, scenarios, audit checks and sensitivity tables.
Python/Pandas	Repeatable SEC extraction, cleaning, period standardization, backtests and error metrics.
SQL (SQLite is sufficient)	Normalized history, unique fact keys, controlled joins, audit queries and reproducible KPI aggregations.
Power BI	Historical vs forecast financial visualizations and scenario-sensitive KPI views.

06 / FORECAST LOGIC

Drivers, scenarios & sensitivities

Forecast business operations, not unexplained spreadsheet numbers.

Forecast approach

Use a base case anchored in historical operating trends and filing disclosures. If segment revenue is clearly available, forecast segments individually; otherwise disclose an aggregate revenue-growth assumption. Forecast expenses and balance-sheet items using appropriate margins, turnover ratios or activity drivers. Record why each assumption is reasonable.

Scenario	Illustrative operating story	Drivers that change
Base	Continuation of defensible recent trends.	Organic growth, margin, capital spending and working capital near normalized historical assumptions.
Upside	Stronger demand and/or operating leverage.	Higher growth, improving margin and/or faster collections, without assuming all drivers improve independently.
Downside	Demand pressure or cost inflation.	Lower growth, margin compression and/or slower collections with realistic financing consequences.

Do not hardcode example growth rates into the finished model. Derive company-specific assumptions from available information; provide a documented judgment and reasonable range.

Minimum sensitivities

Sensitivity	Output management should see
Revenue growth × operating margin	Changes in EBIT, net income and free cash flow across a two-variable matrix.
Working capital × capex	Cash liquidity / FCF exposure and the trade-off between growth and cash investment.
Optional: interest or tax	Interest expense and earnings implications, only if material for the selected company.

Assumptions register

For every adjustable forecast driver, record: variable name, unit, initial value, source/basis, scenario range, owner, last changed date, expected directional effect and rationale. Keep assumptions separate from calculated formulas.

> **Core management question:** Under which plausible assumptions does the business generate durable free cash flow, and which 2–3 operating drivers have the largest effect? The final memo must answer this question from modeled evidence.

07 / QUALITY & VALIDATION

Backtesting, benchmarking & model assurance

A transparent estimate that can fail honestly is stronger than an impressive but unverified accuracy claim.

Backtesting protocol

- Choose historical forecast origins that leave at least three prior fiscal years of inputs where feasible.
- At each origin, use only information publicly available as of that origin; retain a source-as-of record.
- Generate one-year-ahead revenue, EBIT and FCF forecasts using the same logic as the current model.
- Compare actuals to forecasts and to a simple baseline (e.g., prior-year growth or recent historical average).
- Report error distributions and directional misses; explain why misses happened and avoid tuning to the held-out years.

Recommended metrics

Measure	Purpose / caveat
MAE	Average absolute forecast error in original units; easy to interpret.
MAPE	Percentage error when actual values are nonzero and sufficiently stable; misleading near zero or for negative values.
WAPE or sMAPE	Supplementary scaled alternatives when MAPE is unstable, with limitations documented.
Bias	Average signed error; shows chronic overforecasting or underforecasting.
Baseline gap	Compare error against the naive benchmark; report improvement only if observed.

Acceptance tests

Check	Pass requirement
Historical reconciliation	Every material historical statement item ties to cited filings; differences tracked and resolved or disclosed.
Statement linkage	All forecast years balance within stated rounding tolerance, with no unexplained plugs.
Scenario behavior	Downside/upside assumptions propagate consistently; sensitivities update automatically.
Leakage prevention	Backtests use only data available at each forecast origin.
Reproducibility	A reviewer can refresh sources and rerun computations following README steps.

08 / DECISION SUPPORT

Dashboards & executive recommendations

The deliverable must answer “so what?” rather than merely display financial ratios.

Power BI report pages

Page	Recommended views / interactions
01 | Financial snapshot	Historical actuals vs forecast, revenue, EBIT, operating margin, FCF, cash, debt and growth.
02 | Driver analysis	Revenue and profit bridges; margin trend; cost mix; DSO/DPO/DIO as applicable.
03 | Scenarios	Base/upside/downside comparisons, FCF waterfall, liquidity trend and scenario parameter notes.
04 | Accuracy & risks	Backtest forecast-vs-actual, baseline comparison, error, variance drivers and model caveats.

Required dashboard behaviors

- Fiscal-year labels, $ units, sign conventions and metric definitions are visible.
- Scenario selection updates relevant projections without manually editing the dashboard.
- Actual and projected values are visually distinguishable; no forecast is presented as reported history.
- Every chart is tied to a documented data model or calculation.
- Report is navigable from screenshots or a short video if Power BI cannot be opened by the reviewer.

Executive memo: recommended two-page structure

Section	Content
Executive answer	One-paragraph recommendation and 2–3 headline facts supported by the model.
Historical context	Where performance changed, why that matters, and what is uncertain.
Forecast / scenarios	Base-case outlook plus upside/downside range and highest-impact drivers.
Actions and risks	2–4 recommendations with triggers, trade-offs, monitoring metrics and limitations.
Methodology footnote	Company/source, as-of date, forecast horizon, assumption policy and backtest caveats.

> No need to manufacture “saved money” or “improved profitability” outcomes. The demonstrated impact is the quality, traceability and usefulness of the analysis and recommendations.

09 / DELIVERABLES

Required project handoff package

What a reviewer should be able to open, inspect, reproduce, and discuss.

Artifact	Must contain / format
Financial model	Excel .xlsx: raw actuals reference, normalized actuals, assumptions, three statements, scenarios, sensitivities, audit checks.
Data pipeline	Python .py or notebooks, dependency/requirements file, SEC/FRED fetch or import steps, transformations.
Auditable data store	SQLite .db or reproducible .sql scripts; data dictionary, primary keys and data quality checks.
Dashboard	Power BI .pbix plus accessible PDF screenshots / visual export with source and metric notes.
Executive recommendation	Two-page PDF memo describing decision, evidence, scenario outcomes and risks.
Documentation	README, data-source manifest, modeling assumptions register, methods, limitations and demo instructions.
Evaluation evidence	Backtest results CSV/Excel, comparison with baseline, reconciliation log, automated test output.
Portfolio presentation	3–5 minute recorded/screen demo or a 6–8-slide executive walkthrough, plus a GitHub repository.

Suggested repository layout

financial-forecasting-case-study/

```
README.md
docs/ (business_case.pdf, executive_memo.pdf, source_manifest.csv)
data/raw/ (read-only source extracts; note licensing and file size)
data/processed/ (clean history and metrics)
src/ (ingest.py, normalize.py, forecast.py, backtest.py)
tests/ (accounting_identity, periods, units, no_leakage)
models/ (financial_model.xlsx)
dashboards/ (financial_report.pbix, screenshots/)
results/ (forecast_errors.csv, scenario_summary.csv)
```

Minimum viable delivery: One accurate financial model + one backtest + three scenarios + a concise memo. SQL and Power BI are useful extensions, not excuses to postpone a correct model.

10 / EXECUTION PLAN

Workstreams, milestones & resources

A staged plan keeps modeling and analysis quality ahead of cosmetic dashboard work.

Phase	Major activities	Exit deliverable	Indicative effort
1 | Frame	Select company; confirm filings, reporting calendar and management question.	One-page research brief + source plan.	5–8 h
2 | Source & reconcile	Extract five FYs; normalize tags; check statement totals and units.	Sourced history + data dictionary + reconciliation log.	10–15 h
3 | Forecast & validate	Link statements; scenarios; sensitivities; rolling backtests; baseline.	Working .xlsx model + error and test results.	16–23 h
4 | Communicate	Build dashboard; executive memo; presentation and README.	Power BI/report visuals + memo + demo.	10–15 h
5 | QA / package	Peer-style review, repeatability test, polish presentation, archive files.	Versioned final repository + evidence checklist.	4–7 h

Effort estimate: Approximately 45–68 focused hours in total, depending on source complexity and tool familiarity; this is a planning assumption, not a completion promise. Work can be spread over 4–6 weeks at a realistic personal pace.

Dependencies and stage gates

Gate	Proceed only when…
G1 | Data viable	Five years of usable disclosures and a consistent fiscal-year mapping are confirmed.
G2 | Model sound	Historical and forecast accounting checks pass and assumptions are documented.
G3 | Evaluation honest	Holdout periods and naive baseline are fixed before judging forecast errors.
G4 | Portfolio ready	README, memo, audit trail and a working viewer path are complete.

Expected resource needs

- Hardware: standard personal computer with a spreadsheet application, Python and local data storage.
- Software: Excel or compatible workbook editor, Python/Pandas, SQLite, Git/GitHub, Power BI Desktop where supported.
- Data: free SEC filings/APIs; optional FRED macro series, subject to API key and data terms.
- Cash cost: likely low for the MVP using public data and available software, but individual license/subscription costs may apply.

11 / RISKS & CONTROLS

Failure modes and mitigation plan

The most credible finance portfolio projects document what could go wrong and how decisions would change.

Risk	Severity	Control / response
XBRL tag or period mismatch	High	Reconcile to source filings; use a line-item mapping table; separate annual/YTD/quarterly facts.
Statements fail to balance	High	Add systematic identity checks; surface unmatched adjustments; never hide with plugs.
Limited historical sample	High	Prefer interpretable assumptions; compare with naive baseline and report uncertainty; avoid spurious ML accuracy claims.
Hindsight / data leakage	High	Snapshot filing availability by origin date; do not use later-restated facts in earlier-origin tests without disclosure.
Overcomplex tooling	Medium	Complete spreadsheet model first; add SQL/BI only after auditable core exists.
Overfitting assumptions	Medium	Freeze backtest protocol, report misses, and document judgment vs measured evidence.
Misleading presentation	Medium	Label history, forecasts, modeled scenarios and forward-looking uncertainty separately.
Software / API accessibility	Medium	Keep source exports and static PDF images; provide manual-source fallback and reproducible instructions.

Ethical and resume-claim controls

- State that this is an independent public-data analysis, not company-sponsored work or production FP&A.;
- Do not imply audited forecast figures or actual earnings changes caused by this project.
- Only report accuracy, sample sizes, efficiencies or scenario outcomes after producing and recording them.
- Treat results as educational analysis; include “not investment advice” where discussing public company prospects.

Go / no-go decision: If a selected company cannot support a clean source-to-model reconciliation, switch companies rather than silently inventing missing financial data.

12 / FINAL ACCEPTANCE

Definition of done & portfolio readiness

The project is complete only when each evidence gate can be demonstrated from files, not assertions.

Gate	Review evidence
Source integrity	At least five usable fiscal years; 10-K/10-Q source manifest, fiscal calendar, normalized units, reconciliation notes.
Three statements	Income statement, balance sheet and cash flow forecast three years; accounting identities pass.
Drivers / scenarios	Assumption register, base/upside/downside cases and at least two sensitivity analyses.
Backtesting	Defined historical origins, as-of data policy, baseline, errors for revenue/EBIT/FCF and written limitations.
Decision support	Four useful report pages (or equivalent), a two-page memo and 2–4 traceable recommendations.
Reproducibility	Excel formulas, code, raw/processed data documentation, test suite and clear README.
Communication	A screenshot-friendly project view, a short walkthrough, and accurate portfolio/resume wording.

What a strong final resume bullet could eventually say

> **Template (do not use before completion):** “Developed a three-statement financial model from [N] years of public filings, forecasting revenue and free cash flow across three scenarios; backtested against [baseline] and presented [N] management recommendations.” Replace brackets only with measured, reproducible results.

Exit package reviewer test

A reviewer should be able to (1) find the original financial source, (2) understand how the next year is forecast, (3) change an assumption and see all statements reconcile, (4) understand forecast error vs baseline, and (5) read the main recommendation in under one minute.

APPENDIX / REFERENCE

Model formulas, resources & decision log

A compact technical reference to use when implementation begins.

Core relationship	Model treatment
Revenue growth	Revenue_t / Revenue_(t-1) - 1; or sum of segment drivers where disclosed.
Gross profit	Revenue - cost of goods/services sold.
EBIT / operating profit	Gross profit - operating expenses (plus/minus disclosed operating adjustments).
Net income	Pretax income - taxes, after considering interest/nonoperating items.
Net working capital	Operating current assets - operating current liabilities; exclude cash and financing debt from operating NWC.
FCF (common convention)	Cash from operating activities - capital expenditures; disclose alternative conventions if used.
Cash rollforward	Ending cash = beginning cash + CFO + CFI + CFF (+ FX/other reported adjustments where applicable).
BS identity	Assets - liabilities - equity = 0 within documented rounding tolerance.
MAPE	Mean(abs((actual - forecast) / actual)); avoid where actuals are zero/near-zero or negative.
Bias	Mean(forecast - actual); positive values indicate overforecasting under this sign convention.

Authoritative information sources

SEC EDGAR APIs: https://www.sec.gov/search-filings/edgar-application-programming-interfaces
SEC developer / fair access: https://www.sec.gov/about/developer-resources
FRED API overview: https://fred.stlouisfed.org/docs/api/fred/overview.html
FRED observations endpoint: https://fred.stlouisfed.org/docs/api/fred/series_observations.html

Sources checked November 7, 2026. The SEC supports JSON XBRL company facts and filing histories; public API access has technical and fair-access conditions. FRED offers an API for economic time series, and its documented series-observations endpoint requires a registered API key.

First actions

1. Select the company and primary question. **2.** Inspect five years of 10-K filings. **3.** Draft the source manifest and driver tree. **4.** Model reported history before entering any forecast. **5.** Freeze the backtest design before evaluating results.

> **Portfolio framing:** “Corporate Financial Forecasting & Performance Analysis — independent financial modeling project.” Keep the project employer-neutral and finance-first; adapt the resume emphasis to each role after actual evidence exists.